Second-Order Accord Co-occurrence for Interaction-Free Perfume Retrieval

***Abstract*—Perfume recommender systems help customers find products that match a reference scent, and they typically do so by comparing accords, the individual scent notes such as amber or woody, one at a time, treating each accord as independent. This ignores that accords carry meaning in combination, since two products sharing a pair of accords can resemble each other more than their separate accords suggest. The gap is most acute for product catalogs that record no ratings, no purchase history, and no other user behavior. In such an interaction-free setting, the widely used collaborative filtering methods cannot be applied, and accord content becomes the only available signal. Whether modeling how accords combine improves recommendation in this setting, and how far the benefit extends, has not been established. This paper studies the problem on a real catalog of 340 products from an Indonesian perfume brand, where the task is to find the product that most closely resembles a given reference perfume. The proposed method extends TF-IDF, a standard term-weighting technique, so that it scores accord pairs occurring together in a product in addition to individual accords, without any parameter learned from data. It is compared against fourteen established methods, and a procedure is introduced to detect whether the catalog labels leaked into the accord features used to predict them. Scoring accord pairs improves the mean reciprocal rank, which measures how highly the correct product is ranked, from 0.505 to 0.566 ($p < 10^{-7}$), and the improvement stops there, because scoring triples of accords or larger groups adds no further benefit. Adding a trained model on top of the accord-pair representation gives no significant gain, so the simpler parameter-free version is retained. The audit finds that part of the benchmark is derivative, and the same conclusions hold on the subset free of this issue. These results imply that a lightweight and transparent method is sufficient for recommending within catalogs that carry no user data, which makes it deployable in stores where staff face peak-hour queues, and the shared accord pairs behind each recommendation can be surfaced directly as an explanation to the customer.**

***Index Terms*—recommender systems, content-based retrieval, co-occurrence, TF-IDF, cold start, interaction-free, fragrance**

I. Introduction

The task of recommending perfume is a highly personal and subjective one, based on memory, emotion and the intricate interaction between the senses [1], and performed in a fast-growing consumer market [2]. Aromatique is an Indonesian fragrance company that provides locally produced aromas in a growing number of outlets. The sales staff during rush hours bear the burden of memorizing reference fragrances for a customer, and then knowing the closest in-house product off-hand. The brand needs an automated system that, given a reference perfume, finds the catalog products that are the most similar to it.

The main limitation is that the catalog only contains content. Each product is identified by its olfactory accords (*amber*, *woody*, or *warm spicy*) and a family label, with no clicks, no ratings, no purchase history. Therefore, the environment has no interactions and corresponds to an ultra cold-start scenario with small catalog. This rules out the dominant types of recommenders. In the absence of an interaction signal, collaborative filtering is unable to generate predictions, popularly referred to as the new-item cold-start problem [3]. In absence of interactions item attributes can only be matched [4], which means that item cold-start methods have to rely on content and item similarity [5]. In limited datasets [6] lightweight methods are generally more accurate and suitable than high-capacity models.

The quality of retrieval is only dependent on the representation of accords as there is no other signal. Current perfume recommendation systems mainly use first-order methods, in which candidates are ranked according to the similarity between individual accords with cosine similarity of term frequency-inverse document frequency (TF-IDF) vectors or Jaccard similarity of accord sets, assuming each accord is independent [2], [7]. Accords interact, their simultaneous presence reveals information that cannot be expressed by any one accord, which is why linguistic descriptions taken in isolation are insufficient to convey perception [8]. The utility of modeling co-occurrence has been consistently demonstrated in recommendations, e.g., for item sessions [9], [10], item pairings [11], or higher-order group structures [12], however these approaches rely on co-occurrence information from user interactions not available in the interaction-free setting. This paper addresses the problem of modeling the co-occurrence of accords that characterize each product, where accords are the only available signal.

This study uses the real list of 340 items of Aromatique. The paper has three contributions:

- Interaction free perfume retrieval using a second order co-occurrence method.
- Cross-paradigm evaluation of fourteen methods including marginal similarity, neural embeddings, graph embeddings, perceptual taxonomy and learning models in interaction-free setting.
- A process to audit label circularity, measuring the existence of mimic label in accord features, and classifying the evaluation.

Three research questions are posed:

RQ1. Does modeling accord co-occurrence at order two or above improve retrieval over the order-one baseline in the interaction-free regime?

RQ2. How far does the improvement continue with interaction order before it saturates?

RQ3. How does the second-order representation compare with established paradigms?

II. Related Work

A. Perfume recommendation

Most existing scent recommendation systems vary in the type of data they use. Nurmuthia *et al.* [2] rank 24,760 Fragrantica perfumes by matching a user-generated free-text description against note lists using cosine and Jaccard similarity, evaluated with nDCG@3. Aurora and Baizal [7] use content-based filtering based on TF-IDF on perfume reviews and user ratings. TF-IDF is chosen as an effective method to quantify the textual data. Widjaja and Anggreainy [13] perform aspect-based sentiment analysis on about 160,000 reviews of 740 perfumes, evaluating BERT, RoBERTa and DistilBERT to evaluate scent ratings from opinion text. Kalashi *et al.* [14] built a network of fragrance preferences based on user evaluations and detected communities with similar tastes. Lutan *et al.* [1] review the field and highlight the main challenges to be subjectivity and the sensory complexity. Each of these methods requires an external signal to the catalog, e.g. a user generated description, evaluations, ratings or a preference network. The catalog studied here contains none of them, which places it outside the operating conditions of all four systems.

B. Co-occurrence and higher-order structure

It has been shown that co-occurrence encodes information not captured by similarity of individual items. Zhang *et al.* [9] point out that session-based approaches extract limited item co-occurrence patterns indicated by item ID and enrich them with multi-modal information, such as photos, descriptive text, and price. Gwadabe and Liu [10] construct an item co-occurrence graph over sessions, and decompose the ranking score into global, local and co-occurrence parts, each evaluated with MRR. Cai *et al.* [11] connect items via pointwise mutual information over consumption logs, intersecting co-occurring features. Hao *et al.* [12] extend their analysis from pairs to simplices, modelling k-groups across several behaviour patterns. Haddou *et al.* [15] analyse 37 similarity measures that consider negative co-occurrence for implicit ratings and find that no single measure dominates across datasets. Wang *et al.* [16] propagate high-order connectivity using a user-item knowledge graph with attention mechanisms. In all cases the co-occurrence is defined by user interaction records such as sessions, ratings or behavioural logs. None of them estimates co-occurrence among the intrinsic attributes of a single item, which is the only form available when interaction records do not exist.

C. Cold start and small data

The central feature of the cold-start problem is the absence of interactions. Deldjoo *et al.* [3] deal with new items with no interactions, where collaborative filtering cannot make predictions and extract a content genome from the item itself. Bernardis and Cremonesi [5] map item content attributes to a low-dimensional embedding space to emulate collaborative information for items with no past preferences. In the paper of Kuznetsov and Kordík [4] the focus is put on the phase with few interactions. A recommender only matches item attributes, using an ontology to improve diversity. Besides the cold start problem, model capacity itself becomes a disadvantage at this scale: Rather *et al.* [6] study small-data scenarios and find lightweight approaches to be more beneficial, and Li *et al.* [17] and Yao *et al.* [18] also report cases of overfitting when training networks on small, class-imbalanced data. Additionally, Ahlmann-Eltze *et al.* [19] demonstrate that complex models do not outperform simple linear baselines in critical evaluations. This literature argues that content is the only feasible signal without interactions and simple approaches are the right approach at a small scale. It does not address which content representation to use when that content consists of accords.

D. Representing fragrance content without interactions

In the absence of an interaction signal, a fragrance can be characterized only by its intrinsic composition, and three approaches are presented in the previous research. In the lexical route, the authors treat accords as words and rank them based on word overlap, weighted by rarity, as defined in the probabilistic relevance framework and BM25 [20]. This framework is the basis of the TF-IDF ranking used by current perfume recommenders [2], [7]. It considers each term independently and, therefore, does not express the co-occurrence of terms. The dense route encodes text into vectors, where proximity indicates semantic similarity, as in Sentence-BERT [21], or embeds entities from a graph through random walk sampling, as in skip-gram and node2vec [22], [23]; these methods derive their representations from data and require a sufficiently large corpus, while the catalog considered here consists of 340 products and short accord lists. The perceptual route encodes pre-existing knowledge of odor structure rather than acquiring it: Zarzo and Stanton [24] examine the olfactory profiles of 140 perfumes through principal component analysis to create a sensory wheel that categorizes accords into families, while Lee *et al.* [25] develop a principal odor map based on molecular architecture, and Ravia *et al.* [26] establish a perceptual distance between odorants; these provide a fixed taxonomy of accord relationships, which are predetermined rather than derived from the existing catalog. Thus, each route covers accords either in isolation (as the lexical route) or via a similarity framework based on large scale data or defined a priori (as the dense and perceptual routes). None estimates how accords co-occur within the products of the catalog itself, which is the signal this work targets.

III. Methodology

A. Dataset

The list has $M = 340$ Aromatique products. For each product p there is a family label and an accord set $A(p) \subseteq \mathcal{V}$

over an accord vocabulary $\mathcal{V}$. The ground truth is a collection of reference perfumes. Each labeled product is mapped to the branded original it imitates, and this imitation relation is held out for evaluation. Accord lists were normalized to a single vocabulary, using the approach of numerically encoding the Fragrantica-style accord descriptors for analysis of perfumes [24]. Of the 340 products in total, 243 are imitation-labeled and 97 are unlabeled originals as distractors. This results in 210 labeled query-product pairs, and a set of 208 references with at least one labeled product for leave-one-out evaluation. Table I summarizes the dataset.

B. Second-order accord co-occurrence

The crucial measure is the interaction order, i.e. the size of the accord subset used as a scoring token. The formulation uses two ideas from the recommendation literature. Hao *et al.* [12] define higher-order structures as simplices (sets of objects of size k instead of pairs) and show that raising k captures group-level regularities that pairwise connections miss. Gwadabe and Liu [10] build an item co-occurrence graph and decompose the ranking score into different components, each of which is assessed with MRR, thus making the contribution of the co-occurrence term explicit rather than implicit. Both derive their structure from user interaction logs, which are unavailable in this setting. The study applies the two concepts on the content of items, so that the size of the subset k can vary for the different accords of a single product, and the score can be decomposed by k, so that the contribution of each order can be measured directly. The order-k tokens of a fragrance f are its k-sized subsets of accords,

$$T_k(f) = \{\, S \subseteq A(f) \;:\; |S| = k \,\}, \tag{1}$$

so order one is the single accords that give marginal similarity and order two the co-occurring pairs. A representation of order N pools all tokens of order $1 \le k \le N$,

$$T_{\le N}(f) = \bigcup_{k=1}^{N} T_k(f). \tag{2}$$

Order one returns the individual scores of a plain TF-IDF baseline, order two adds the co-occurring pairs provided by this study. This item-content centric notion of order is analogous to the high-order connectivity exploited by graph-based recommenders from interactions [10], [16], which is here computed based on the internal item accord structure.

TABLE I
DATASET STATISTICS

Quantity	Value
Products (catalog pool)	340
labeled / unlabeled distractors	243 / 97
Reference queries (leave-one-out)	208
Labeled query–product pairs	210
Accord vocabulary $\lvert\mathcal{V}\rvert$	59
shared product/query accords	55

Each token is weighted by inverse document frequency over the $M = 340$ products, a standard rarity weighting from the probabilistic relevance framework [20], with document frequency $\mathrm{df}(t) = |\{f : t \in T_{\le N}(f)\}|$,

$$\mathrm{idf}(t) = \log \frac{1+M}{1+\mathrm{df}(t)} + 1. \tag{3}$$

Rare co-occurring pairs receive a high weight and ubiquitous single accords a low one, so the hub penalty is automatic. A fragrance becomes a sparse vector v_f with $v_f[t] = \mathrm{idf}(t)$ for $t \in T_{\le N}(f)$ and 0 otherwise, then L_2-normalized to $\hat{v}_f = v_f / \|v_f\|_2$. The score of a product p for a reference q is the cosine of the normalized vectors, which separates additively into per-order contributions,

$$s(q,p) = \langle \hat{v}_q, \hat{v}_p \rangle = \sum_{k=1}^{N} S_k(q,p), \tag{4}$$

where the order-k term aggregates the shared order-k tokens,

$$S_k(q,p) = \frac{1}{\|v_q\|\,\|v_p\|} \sum_{t \in T_k(q) \cap T_k(p)} \mathrm{idf}(t)^2. \tag{5}$$

The recommendations are generated by sorting the 340 products by $s(q,p)$. The matcher has no learned parameters, and the sole design choice is the maximum order N, a hyperparameter chosen via cross-validation (Section IV-B). Fig. 1 shows the pipeline, and Fig. 2 shows the difference between first order and second order. The second-order representation shows the common accord pairs of each recommendation as verifiable co-occurrence routes for a subsequent explanation layer, in line with path-based explainable recommendation [27]. These paths are reported as an artifact rather than a faithfulness claim.

C. Evaluation protocol

Each method scores all 340 products for a reference q and is judged by the rank r_q of the ground-truth product. It is not a classification task but a retrieval task, since for each query

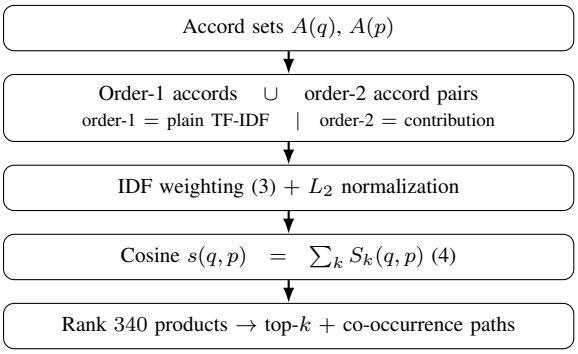


Fig. 1. Scoring pipeline of the proposed second-order matcher, showing the expansion of accord sets into order-one and second-order tokens, IDF weighting and normalization, and cosine ranking of the catalog.

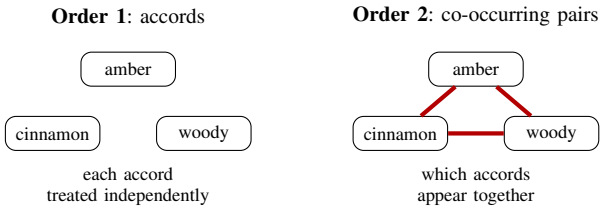


Fig. 2. Illustration of the two interaction orders, contrasting the single-accord representation of order one with the co-occurring accord pairs added at order two.

there is exactly one relevant product. Quality is measured in terms of mean reciprocal rank (MRR) and Hits@k,

$$\mathrm{MRR} = \frac{1}{|\mathcal{Q}|}\sum_{q}\frac{1}{r_q}, \qquad \mathrm{Hits@}k = \frac{1}{|\mathcal{Q}|}\sum_{q}\mathbf{1}[\,r_q \le k\,]. \tag{6}$$

Metrics such as Precision, Recall, F1 and Accuracy are not suitable as they require a binary relevance threshold and ignore ordering. Therefore, a relevant product ranked second would get the same score as a two-hundredth one, while MRR and Hits@k properly reward the ranking of the single correct answer. Hits@3 is of immediate and practical value, since the deployed system returns the three highest ranked prospects. This work adopts Mean Reciprocal Rank (MRR) as the standard evaluation metric for co-occurrence-augmented retrieval [10], and the lack of a single dominant similarity metric across datasets [15] motivates evaluating multiple comparators rather than one.

All methods are evaluated on the full set of 340 products under leave-one-out over the 208 queries. Learned rankers are scored out-of-fold with 5-fold group cross-validation grouped by query, so a query never appears in both training and test, and stochastic methods are averaged over five seeds (mean±std). Significance uses the Wilcoxon signed-rank test paired across queries on reciprocal ranks with Holm correction, together with 95% confidence intervals from 10,000 bootstrap resamples. A difference is called significant only when the Holm-adjusted $p < 0.05$, the bootstrap interval excludes zero, and $|\Delta\mathrm{MRR}| \ge 0.01$.

D. Label-circularity audit

The main problem of validity is circularity, because the accords of a product that is copied from the reference perfume it emulates will partially replicate the label because of overlap in the accords. Since Fragrantica shows accords in a fixed order, transcription would preserve this order. The audit compares the number of times $A(p)$ is an ordered subsequence or an ordered prefix of $A(q)$, for each labeled pair to a null hypothesis based on 1,000 random permutations of the query accord order. Queries whose imitation shows ordered containment are flagged, and results are reported both on the full set (ALL) and on the subset with no evidence of ordered transcription (NON-OP).

IV. Results and Discussion

A. Dataset and circularity audit

The audit finds ordered structure far beyond chance. Among 210 labeled pairs, 89 have the product accords as an exact ordered prefix of the query accords and 100 as an ordered subsequence, against null means of 0.2 and 3.4 respectively (empirical $p = 0.001$). The verdict is that the benchmark is partially derivative, because a substantial share of pairs is consistent with transcription from the reference. Accordingly the full set (ALL, 208 queries) is reported alongside the subset without evidence of ordered transcription (NON-OP, 109 queries), and the latter is treated as the primary evaluation. The NON-OP subset is not a certified-clean set but one showing no ordered-transcription signature, and transcription that reordered accords would not be detected.

B. RQ1 and RQ2: co-occurrence helps and then saturates

Adding second-order co-occurrence tokens raises accuracy sharply and then plateaus (Fig. 3). On the full set MRR rises from 0.505 at order one to 0.566 at order two, a gain of $\Delta\mathrm{MRR} = +0.061$ that is strongly significant, with $p_{\mathrm{adj}} = 1.2 \times 10^{-7}$ and 95% CI $[+0.034, +0.090]$. On the primary NON-OP subset the gain is $+0.067$ ($p_{\mathrm{adj}} = 7.5\times10^{-4}$), which confirms RQ1. The improvement then saturates at order two. Order three is not significantly better than order two on either set, with $\Delta\mathrm{MRR} = +0.002$ ($p_{\mathrm{adj}} = 1.00$) on ALL and $\Delta\mathrm{MRR} = -0.004$ ($p_{\mathrm{adj}} = 0.07$) on NON-OP, and higher orders add nothing. The saturation point is therefore $N^* = 2$ under the decision rule of Section III-C. A nested cross-validation that selects N to maximize training MRR tends to pick a higher order, but the selected order confers no significant test-set advantage over order two, so both outcomes are reported rather than choosing the more favorable one. This answers RQ2, because the co-occurrence signal is captured at order two and modeling triples adds no robust benefit on this catalog.

C. RQ3: comparison with established paradigms

Table II compares the second-order matcher (P1) with the comparators across five paradigms, and two readings stand out. First, methods that build a representation over accord pairs occupy the top of the table. The proposed second-order matcher and its learned-fusion variant P2 lead, with the signature ranker A3 and the PPMI–SVD embedding A2 close behind, while methods that never form pair features, namely Jaccard, BM25, the wheel transport A1, and the graph embeddings, trail. This is consistent with RQ1, because the pair signal, however it is encoded, is what separates the strong methods from the weak ones. Second, exact unweighted co-occurrence is not improved by learning it. P2 augments the second-order representation with a logistic fusion trained on labels, yet on the primary NON-OP subset it is statistically indistinguishable from the parameter-free P1, with $\Delta\mathrm{MRR} = +0.002$ and a confidence interval spanning zero. Because the fusion adds trained parameters without a significant gain, the parameter-free second-order form is retained as the method of record,

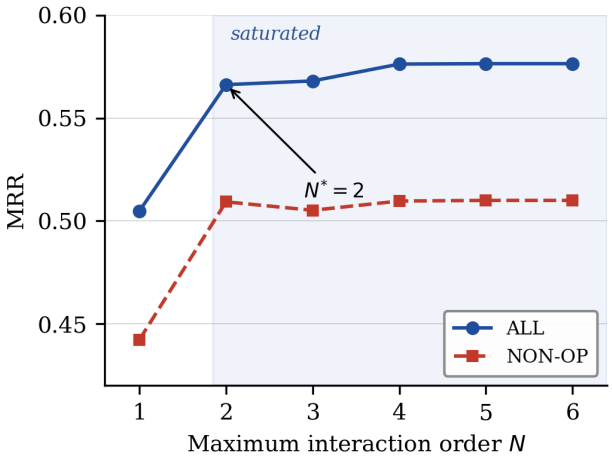


Fig. 3. Retrieval accuracy as a function of maximum interaction order, on the full set (ALL) and the subset without ordered-transcription evidence (NON-OP). The second-order point is the adopted operating order ($N^* = 2$), beyond which the gain is not statistically significant.

TABLE II
RETRIEVAL ACCURACY OF THE PROPOSED SECOND-ORDER MATCHER AND FOURTEEN COMPARATORS ACROSS FIVE PARADIGMS, ON THE FULL SET AND THE DERIVATIVE-FREE SUBSET. P2 DENOTES THE LEARNED-FUSION VARIANT OF THE PROPOSED METHOD, AND BOLD MARKS THE METHOD OF RECORD.

		ALL		NON-OP	
ID	**Method**	**MRR**	**H@3**	**MRR**	**H@3**
P1	**order-2 co-occurrence**	**0.566**	**0.649**	**0.509**	**0.569**
P2	order-2 + learned fusion	0.567	0.647	0.511	0.562
Marginal similarity					
B1	Jaccard	0.351	0.394	0.318	0.330
B2	TF-IDF cosine	0.505	0.582	0.442	0.495
B3	BM25	0.416	0.495	0.380	0.450
Neural / graph embedding					
B4	Sentence-BERT	0.551	0.601	0.414	0.450
B5	Word2Vec	0.214	0.212	0.176	0.165
B6	node2vec	0.231	0.237	0.186	0.182
Perceptual taxonomy					
A1	Edwards tree-W.	0.303	0.337	0.266	0.312
Learned					
A2	PPMI–SVD	0.505	0.577	0.485	0.523
A3	signature LTR	0.555	0.632	0.492	0.543
A4	bigram salience	0.222	0.245	0.212	0.233
A5	bilinear metric	0.292	0.324	0.245	0.262
A6	GBM fusion	0.461	0.550	0.424	0.488
P3	order-2 hub-IDF	0.516	0.591	0.461	0.505

which mirrors evidence that on small and imbalanced data simple models are not surpassed by more complex ones [6], [17], [18], [19]. The two learned models that try hardest to fit pair weights, per-bigram salience (A4) and the bilinear metric (A5), overfit and fall to the bottom under the one-positive-per-query imbalance.

D. Where retrieval succeeds and fails

Stratifying by how much of the product's accord set is contained in the query reveals a sharp split (Table III). When

TABLE III
RETRIEVAL ACCURACY OF THE PROPOSED SECOND-ORDER MATCHER BY ACCORD-CONTAINMENT STRATUM, WITH THE NUMBER OF QUERIES IN EACH STRATUM.

	prefix	**subseq**	**contained**	**partial**
n	88	11	72	37
MRR	0.629	0.631	0.699	0.140

the product accords are contained in the query, order two reaches an MRR of 0.63 to 0.70, whereas in the partial stratum, where they are not contained, it collapses to 0.14. The aggregate figure averages these two regimes. The partial stratum is where every method fails, because when a product accord is absent from the query there is no lexical or co-occurrence bridge to it.

E. Is the partial stratum improvable?

Whether the partial stratum reflects a fixable modeling gap was tested by auditing its 38 pairs with an accord-affinity matrix built from product co-occurrence alone and no labels. Only 7 pairs are soft-matchable, meaning a missing accord has a close co-occurrence neighbor in the query. A further 17 are likely mislinked, because the product accords fully match a different reference, which indicates a label error, and 14 are genuinely far, because the missing accords have no neighbor and no lexical or co-occurrence method can bridge them. Since the soft-matchable niche is small and dominated by label noise and incoherent accords, a soft-matching extension was not pursued, and the partial-stratum ceiling is structural rather than a limitation of the second-order model. The parameter-free second-order matcher is therefore retained as the final method.

V. CONCLUSION

This work studied the problem of interaction-free perfume retrieval over a real catalog of 340 products, and investigated if modeling the accord co-occurrence improves the performance of the retrieval over the dominant single-accord similarity approach in the field. Modeling co-occurrence at order two significantly improves the mean reciprocal rank compared to the order-one baseline, from 0.505 to 0.566 ($p < 10^{-7}$, and $+0.067$ improvement on the principal derivative-free subset). The gain stabilizes at order two since order three adds no substantial improvement (RQ1, RQ2). Across the paradigms compared, the methods that represent accord pairs lead, and the parameter-free second-order matcher matches its best learned-fusion variant without any trained parameters (RQ3). A label-circularity audit reveals part of the benchmark is derivative, so results are given for both the derivative-free subset and the whole set.

The result is given with its limits preserved. All data are from a single proprietary database in one market and external validity is not established. Accords are binary and have no per-accord intensity, making products with the same accord sets indistinguishable. The imitation connection is a perceptual proxy. Its high query–target accord overlap imposes

a task-intrinsic ceiling, and part of that overlap is consistent with transcription of the reference accords, which the audit flags but cannot fully rule out. The mapping of a small number of accords onto the perceptual wheel used by one comparator is a taxonomic judgement. Other logical steps include intensity weighted co-occurrence, replication over more catalogs and integration of the exported co-occurrence paths into an explanatory framework using path-based reasoning on knowledge graphs [16], [27], [28], [29].

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